

INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2
 in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

CHEMISTRY

9647/02

Paper 2 Structured Questions

18 September 2013

2 hours

Candidates answer on the Question Paper
 Additional Materials: *Data Booklet*

READ THESE INSTRUCTIONS FIRST

Write your index number, name and civics group.
 Write in dark blue or black pen.
 You may use pencil for any diagrams, graphs or rough working.
 Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the space provided.
 A Data Booklet is provided.

You are advised to show all working in calculations.
 You are reminded of the need for good English and
 clear presentation in your answers.
 You are reminded of the need for good handwriting.
 Your final answers should be in 3 significant figures.

You may use a calculator.

The number of marks is given in brackets [] at the end of each
 question or part question.

At the end of the examination, fasten all your work
 securely together.

For Examiner's Use	
1	12
2	15
3	12
4	9
5	13
6	11
Significant figures	
Handwriting	
Total	72

This document consists of **18** printed pages.



Answer ALL questions on the spaces provided.

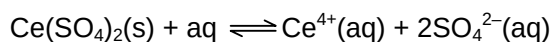
Planning (P)

- 1 The solubility of cerium(IV) sulfate, at a particular temperature, is defined as the mass of cerium(IV) sulfate that will dissolve in and just saturate 100 g of solvent at that temperature. It can also be represented by the following equation:

$$\text{Solubility} = \frac{\text{mass of cerium(IV) sulfate (g)}}{100 \text{ g of solvent}}$$

A **saturated** solution is one in which no more solid can dissolve at a particular temperature.

When cerium(IV) sulfate dissolves in water, the saturated solution is in contact with the undissolved solid to establish the following equilibrium.



Cerium(IV) sulfate is unusual. Its solubility **decreases** as the temperature of the solution increases.

Cerium(IV) sulfate crystals also dissolve in dilute sulfuric acid, H_2SO_4 , a corrosive aqueous solution.

You are to plan an experiment to investigate how the **solubility** of cerium(IV) sulfate crystals in dilute sulfuric acid depends on the concentration of the acid.

- (a) By considering the ions present in a solution of cerium(IV) sulfate in sulfuric acid, predict and explain how the solubility of the cerium(IV) sulfate will be affected by the concentration of the acid.

Prediction: The solubility of cerium sulphate decreases

Explanation: Due to the common ion effect

OR

By Le Chatelier's Principle, as the concentration of sulphuric acid increases, the equilibrium position shift left, suppressing solubility of cerium(IV) sulfate.

- (b) You are to design a two-part laboratory experiment to investigate your prediction in (a).

Part 1 – solubility of cerium(IV) sulfate in water

In addition to the standard apparatus present in a laboratory you are provided with the following materials.

100 cm³ distilled water
solid cerium(IV) sulfate, Ce(SO₄)₂

Give a step-by-step description of how you would

- prepare a saturated solution using all of the 100 cm³ of distilled water,
- control the temperature of the experiment,
- separate the saturated solution from undissolved solid,
- obtain the mass of cerium(IV) sulfate and mass of water in the saturated solution,
- calculate the solubility of cerium(IV) sulfate from the experimental results.

Procedure for Preparation of Saturated solution

1. Using a 100 cm³ measuring cylinder, place 100 cm³ of distilled water into a pre-weighed 250 cm³ dry conical flask/beaker. Weigh again with the distilled water inside to obtain the mass of the distilled water.
2. Maintain the temperature of the water by using a water bath.
3. (Using a spatula, add a large spatula of) Add (solid) Ce(SO₄)₂ into the flask and stir with a glass rod). Continue to add more solid and swirl, until some Ce(SO₄)₂ remains undissolved to ensure a saturated solution is obtained.
4. Filter the (saturated) solution using a dry filter funnel and dry filter paper into a pre-weighed dry conical flask/beaker to ensure the saturated solution is not diluted.
5. Weigh the saturated solution (filtrate), and subtract away the mass of 60cm³ of distilled water to obtain the mass of Ce(SO₄)₂ dissolved in the saturated solution.
6. Using the following equation, the solubility of Ce(SO₄)₂ can be obtained.

$$\text{Solubility} = \frac{\text{mass of solid (g)}}{100 \text{ g of solvent}}$$

Part 2 – solubility of cerium(IV) sulfate in sulfuric acid

In addition to other standard apparatus present in a laboratory, you are provided with the following materials.

100 cm³ volumetric flask
5 mol dm⁻³ sulfuric acid
distilled water
solid cerium(IV) sulfate

Give a step-by-step description of how you would prepare a range of solutions of sulfuric acid of different concentrations. These solutions could then be used to investigate the solubility of cerium(IV) sulfate in different concentrations of sulfuric

acid. See part (c).

Your plan should include details of the following.

- (i) the number and concentrations of the solutions to be prepared
- (ii) the volumes of acid and water used to prepare the solutions
- (iii) the apparatus and method you would use in their preparation

You should include a table to tabulate your details for (i) and (ii).

In (i), you should prepare 5 solutions with different concentrations, and the range should cover at least a fourfold increase in concentration and at least one should be greater than 2.5 mol dm^{-3} . 5 mol dm^{-3} could be one of these.

- 1) Using a burette / pipette (measuring cylinder not accepted), measure 20 cm^3 (can be other volumes in the table) of the 5 mol dm^{-3} sample of H_2SO_4 into a 100 cm^3 volumetric flask.
- 2) Top up to the mark with distilled water.
- 3) Insert the stopper and shake to obtain a homogenous solution
- 4) This will give a stock solution of 1 mol dm^{-3} H_2SO_4 .
- 5) Repeat steps 1 to 3 using the volumes of H_2SO_4 and distilled water specified in the table below:

Expt	Vol of 5 mol dm^{-3} H_2SO_4 / cm^3	Vol of H_2O / cm^3	New conc of H_2SO_4 / mol dm^{-3}
1	20	80	1
2	40	60	2
3	60	40	3
4	80	20	4
5	100	0	5

- (c) Draw another table with appropriate headings to show the data you would record when investigating the solubility of cerium(IV) sulfate in the solutions of sulfuric acid planned in (b), Part 2.

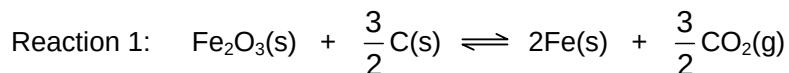
Expt	Vol of 5 mol dm^{-3} H_2SO_4 / cm^3	Vol of H_2O / cm^3	New conc of H_2SO_4 / mol dm^{-3}	M , Mass of original solution [H_2SO_4 and H_2O mixed] / g	Mass of saturated solution [solution + max amt of $\text{Ce}(\text{SO}_4)_2$ dissolved] / g	a , Mass of $\text{Ce}(\text{SO}_4)_2$ dissolved / g	a / M , Solubility of $\text{Ce}(\text{SO}_4)_2$ / g
1	20	80	1				
2	40	60	2				
3	60	40	3				
4	80	20	4				
5	100	0	5				

- (d) State a risk that must be taken into consideration when planning the experiment.

Corrosive nature of sulphuric acid identified as the hazard or any risk associated with this hazard. Alternatively, accept reference to the toxicity of cerium compounds or any risk associated with the use of this reagent.

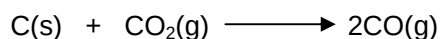
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- 2 (a) Iron is commonly produced in a blast furnace from the ore haematite, which contains iron(III) oxide. Either carbon or carbon monoxide can bring about the reduction of iron(III) oxide.

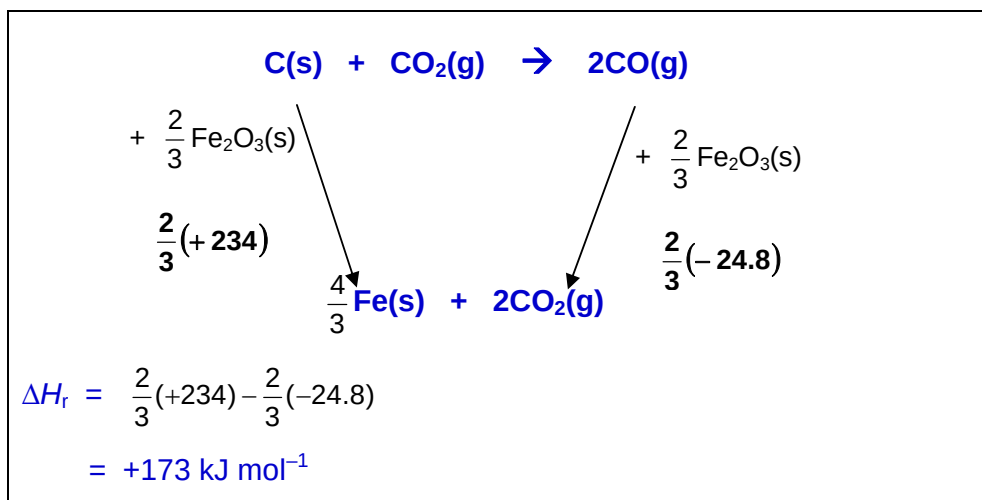


	$\Delta H / \text{kJ mol}^{-1}$	$\Delta S / \text{J mol}^{-1} \text{K}^{-1}$
Reaction 1	+234	+279
Reaction 2	-24.8	+15.2

- (i) The carbon monoxide is formed by the following reaction.



By means of a suitable energy cycle or any other method, calculate the ΔH_r for this reaction using the ΔH data provided for Reactions 1 and 2.



- (ii) Calculate the ΔG of reaction 1 and 2 and state which reaction is more spontaneous at 1800 K.

Reaction 1:
 $\Delta G = \Delta H - T\Delta S$
 $= +234 - (1800)(+279 \times 10^{-3})$
 $= \underline{-268 \text{ kJ mol}^{-1}}$

Reaction 2:
 $\Delta G = \Delta H - T\Delta S$
 $= (-24.8) - (1800)(+15.2 \times 10^{-3})$
 $= \underline{-52.2 \text{ kJ mol}^{-1}}$

Reaction 1 is more energetically feasible at 1800 K.

- (iii) In reality, reaction 2 occurs more readily than reaction 1 at 1800 K. Suggest a reason for this.

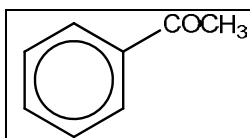
- 1) Reaction 2 is kinetically more favourable as iron(III) oxide reacts more readily with gaseous CO than with solid carbon where reaction can only take place at the interface of the two solids.
- 2) There is not enough energy to overcome the high activation energy of reaction 1 due to breaking of strong C–C covalent bonds in solid carbon for reaction to take place.

[7]

- (b) Iron and its compounds are also commonly used in organic reactions. One such reaction is the Friedel-Crafts acylation reaction. In this reaction, benzene reacts with acyl chlorides, RCOCl , in the presence of iron(III) chloride as a catalyst.

The first step of the mechanism produces the electrophile $\text{R}-\overset{\oplus}{\text{C}}=\text{O}$ that can then attack the benzene ring.

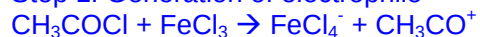
- (i) Draw the product formed when benzene reacts with ethanoyl chloride in the presence of FeCl_3 .



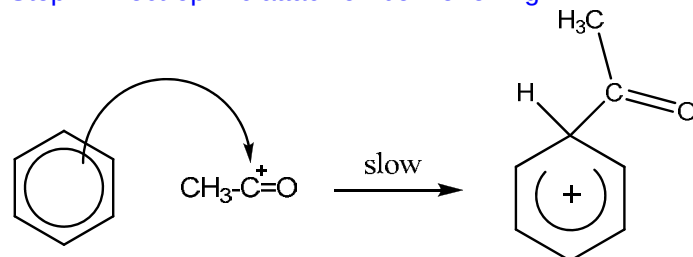
- (ii) Describe the mechanism of the above reaction, including the formation of the electrophile.

Name of mechanism: Electrophilic Substitution

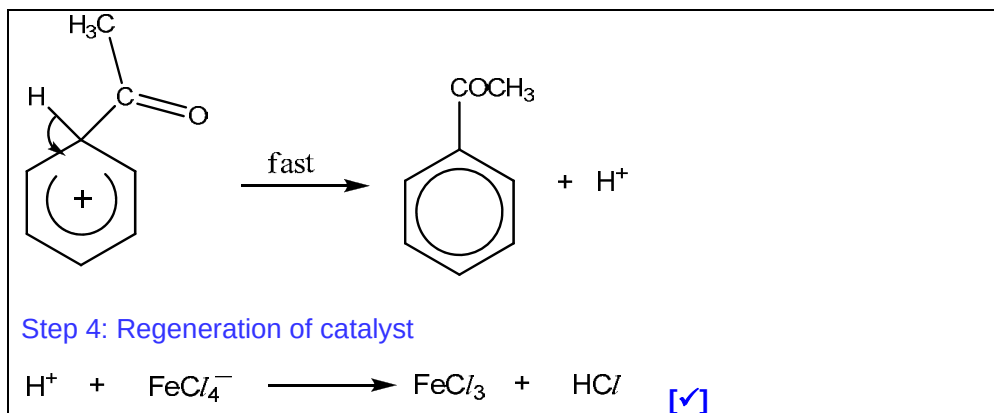
Step 1: Generation of electrophile



Step 2: Electrophilic attack on benzene ring

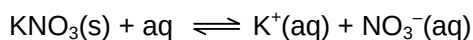


Step 3: Loss of proton



[4]

- (c) (i) The dissolution of potassium nitrate in water is represented by the following equilibrium.



With the aid of the above equilibrium, explain the following phenomenon.

When potassium nitrate dissolves in water, the temperature drops. If a saturated solution of potassium nitrate at 100 °C is cooled to room temperature, crystals are formed.

When potassium nitrate is dissolved, it is an endothermic reaction (OR forward reaction is endothermic) as the temperature drops. This implies that the backward reaction is an exothermic reaction.

On cooling a saturated solution from 100°C to room temperature, heat is removed from the system (OR temperature is decreased). By LCP, the system wants to increase the temperature by giving out heat. Thus, the exothermic reaction is favoured. The position of equilibrium shifts to the left and hence, crystals are formed.

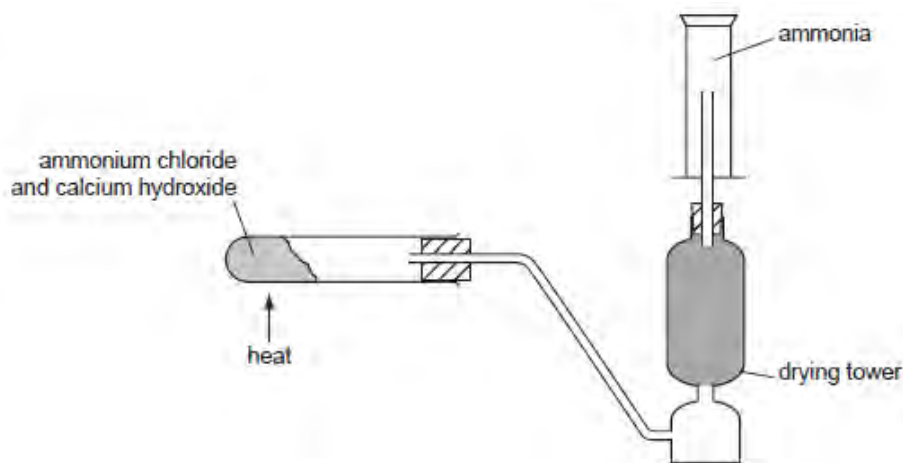
- (ii) Write the K_c expression for the dissolution of potassium nitrate, including its units.

$$K_c = [\text{K}^+][\text{NO}_3^-] \text{ mol}^2 \text{ dm}^{-6}$$

[4]

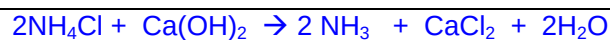
[Total: 15]

- 3 (a) Ammonia may be prepared in a school laboratory by using the apparatus below.



The reaction involves the displacement of ammonia from one of its compounds.

- (i) Construct a balanced equation for the reaction between the two reagents.



Or



- (ii) Common drying agents include calcium oxide, concentrated sulfuric acid and phosphorus(V) oxide.

Which of these could be used in the drying tower in this experiment? Explain your answer.

CaO.

Because it is a base & will not react with NH_3 .

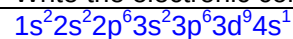
OR

Both P_2O_5 and conc sulfuric acid are acidic & will react with NH_3 .

[3]

- (b) When water is added to white anhydrous CuSO_4 , the solid dissolves to give a blue solution. On addition of concentrated NH_4Cl (aq), the solution changes to a yellow-green due to formation of copper containing species **D**. Concentrating the solution produces green crystals **E** of an ammonium salt with empirical formula $\text{CuN}_2\text{H}_8\text{Cl}_4$.

- (i) Write the electronic configuration of the copper atom.



- (ii) Suggest the formulae of the cation present in **E**.

Cation present in **E**

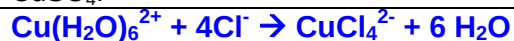


- (iii) Suggest the formulae of anion **D**.

Anion **D**



- (iv) Suggest a balanced equation for the formation of anion **D** from aqueous CuSO_4 .



When excess of $\text{NH}_3(\text{aq})$ is added to species **D**, the yellow green solution turns to a deep blue solution.

- (v) Use this information and the information above to suggest the strength of NH_3 , H_2O and Cl^- ligands in decreasing order.



[5]

- (c) (i) Describe what you would see when aqueous silver nitrate is added to separate tubes containing aqueous sodium chloride, aqueous sodium bromide and aqueous sodium iodide.

white precipitate AgCl

cream precipitate AgBr

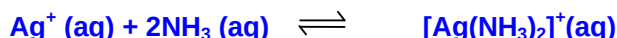
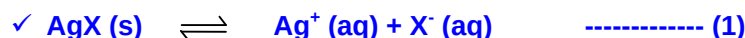
yellow precipitate AgI

- (ii) With reference to (c)(i), describe how the relative solubilities in $\text{NH}_3(\text{aq})$ can be used to distinguish between solutions of the halide ions Cl^- , Br^- , I^- .

✓ AgCl , dissolves (readily) in aqueous NH_3 to give a colourless solution.

✓ AgBr & AgI are insoluble in aqueous NH_3

The decrease in solubility of the three silver halides in ammonia is due to the difference in their K_{sp} values.



✓ When dilute NH_3 is added, Ag^+ will react to form a soluble complex, $[\text{Ag}(\text{NH}_3)_2]^+$. This lowers the $[\text{Ag}^+ (\text{aq})]$ and causes the equilibrium (1) to shift to the right. AgX becomes more soluble.

✓ As the K_{sp} value of AgBr and AgI are lower, the ionic product of AgX is higher than its K_{sp} value. Thus, AgBr and AgI are insoluble in dilute NH_3 . (High concentration of NH_3 is required to shift the equilibrium to the right.)

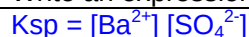
✓ For AgCl , the K_{sp} value of AgCl is higher, thus the ionic product of AgCl is lower than its K_{sp} value. AgCl is soluble in dilute NH_3 .

[4]

[Total: 12]

- 4 (a) The solubility product, K_{sp} , of barium sulfate has a numerical value of $1.08 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$.

(i) Write an expression for the K_{sp} of barium sulfate.



(ii) Use the value of K_{sp} given to calculate the concentration of barium sulfate in a saturated solution.

$$K_{sp} = [\text{Ba}^{2+}] [\text{SO}_4^{2-}] = 1.08 \times 10^{-10}$$

$$[\text{BaSO}_4] = \sqrt{(1.08 \times 10^{-10})} = \underline{1.04 \times 10^{-5} \text{ mol dm}^{-3}}$$

[2]

- (b) Data concerning the elements of Group II of the Periodic Table, at 298K, are given in the table. Further data can be found in the *Data Booklet*.

	Mg	Ca	Sr	Ba
$\Delta H_{\text{hydration}}$ of $\text{M}^{2+} / \text{kJ mol}^{-1}$	−1980	−1650	−1480	−1365
solubility of sulfate/ mol per 100 g of water	1.8×10^{-1}	4.7×10^{-3}	7.1×10^{-5}	9.4×10^{-7}

- (i) When a salt such as a Group II sulfate dissolves in water, the lattice energy must be overcome. Suggest and explain how the magnitude of the lattice energy of Group II sulfates change from MgSO_4 to BaSO_4 . No calculation is required.

According to Coulomb's Law,

$$| \text{Lattice energy} | \propto \left| \frac{q_+ q_-}{r_+ + r_-} \right|$$

where q_+ and q_- : charge on cation and anion respectively

$r_+ + r_-$: sum of radius of cation and anion

Down the group,

- charges on cation and anion remains the same,
 - radius of cation (r_+) increases,
 - However, since the radius of sulfate anion, (r_-) is much larger than the radius of metal cation (r_+),
- ($r_+ + r_-$) remain almost constant, LE is effectively constant

(ii) Suggest why $\Delta H_{\text{hydration}}$ of M^{2+} becomes less negative from Mg^{2+} to Ba^{2+} .

$$\Delta H_{\text{hydration}} \propto \left| \frac{q^+}{r^+} \right|$$

Down the group,

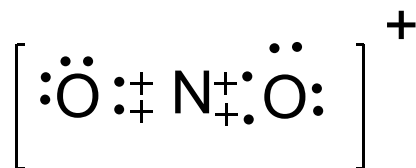
- Charge of cation remains the same
- size of cation increases
- charge density decreases hence form weaker ion-dipole interaction with water

Hence ΔH_{hyd} is less exothermic.

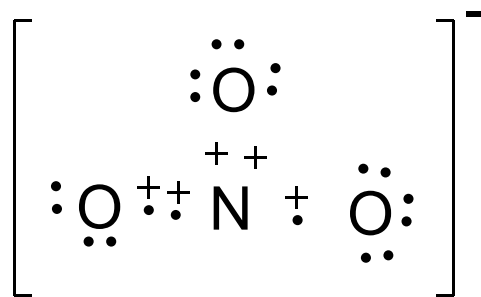
[4]

(c) Dinitrogen pentoxide, N_2O_5 , exist as covalent molecules in the gas phase, but as ionic $[NO_2^+ NO_3^-]$ in the solid or in solution.

Draw dot-and-cross diagrams showing the electrons (outer shells only) in NO_2^+ and NO_3^- , and predict their shapes.



✓ shape: **linear**



✓ shape: **trigonal planar**

[Total: 9]

- 5 Compound **W** has the molecular formula $C_8H_{11}NO$. It is an aromatic compound which contains two functional groups.

Data about the reactions of **W** are given in the table.

reaction	reagent	result
1	Na	colourless gas evolved; white solid formed which is soluble in H_2O
2	$PCl_5(s)$	steamy white fumes
3	alkaline $I_2(aq)$	pale yellow ppt
4	$HCl(aq)$ at room temperature	colourless solution formed
5	$Br_2(aq)$ in excess	white solid formed which has $M_r = 294.8$
6	acidified $KMnO_4$ and heat	purple $KMnO_4$ decolourises

In this question, when identifying functional groups, your answers should be unambiguous.

- (a) (i) Based on reaction **1**, give the names of two different functional groups that could be present in **W**.

Phenol and alcohol

- (ii) Which of the functional groups you have named in (a)(i) is confirmed by reaction **2**? Explain your answer.

Alcohol

Phenol does not undergo substitution with $PCl_5(s)$ as the **C–O bond is strengthened** by the **delocalisation of the lone pair of electrons on O into the benzene ring** OR p orbital of O overlap with π orbitals of benzene.

[4]

- (b) (i) What type of reaction takes place in reaction **3**?

(Mild) oxidation

- (ii) State, as fully as you can, the functional group that reaction **3** shows to be present in **W**.

Alcohol with **$CH_3CH(OH)$ group**

[2]

- (c) (i) What type of reaction takes place in reaction **4**?

Acid base/neutralisation

- (ii) Deduce the molecular formula of the white solid formed in reaction **5**.

$C_8H_9NOBr_2$

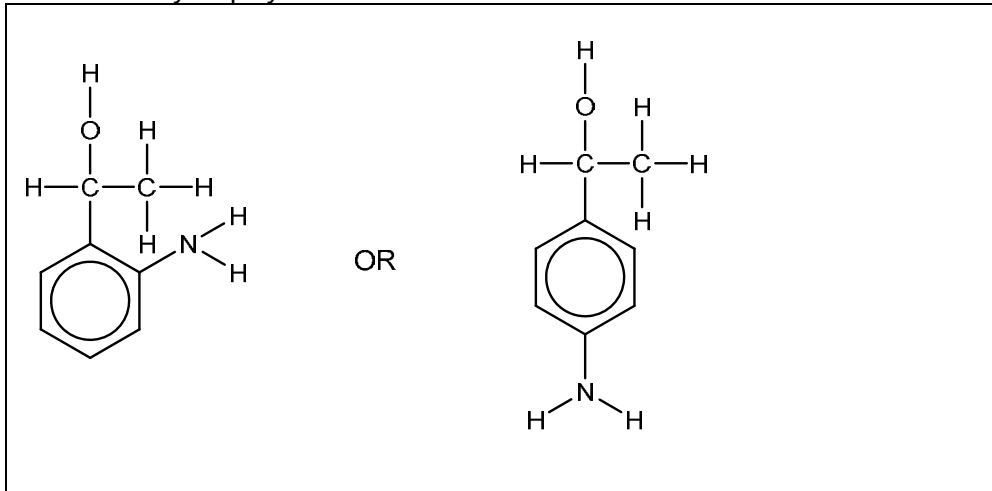
- (iii) Give the functional group that reactions 4 and 5 show to be present in **W**.

Aromatic amine / phenylamine

[3]

- (d) You now have enough information to determine the structural formula of **W**.

- (i) Draw the fully displayed structure of **W**.

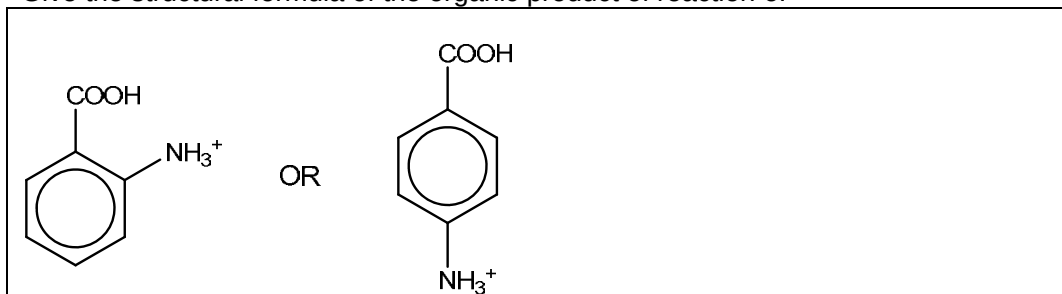


- (ii) Explain clearly why you have placed **each** of the two functional groups in their particular positions.

The -NH_2 group is a strong activating group and is 2,4-directing. From reaction 5, only 2 Br atoms are introduced into the benzene ring during electrophilic substitution. Hence, the alcohol group must be at the 2nd or 4th position with respect to the -NH_2 group.

[3]

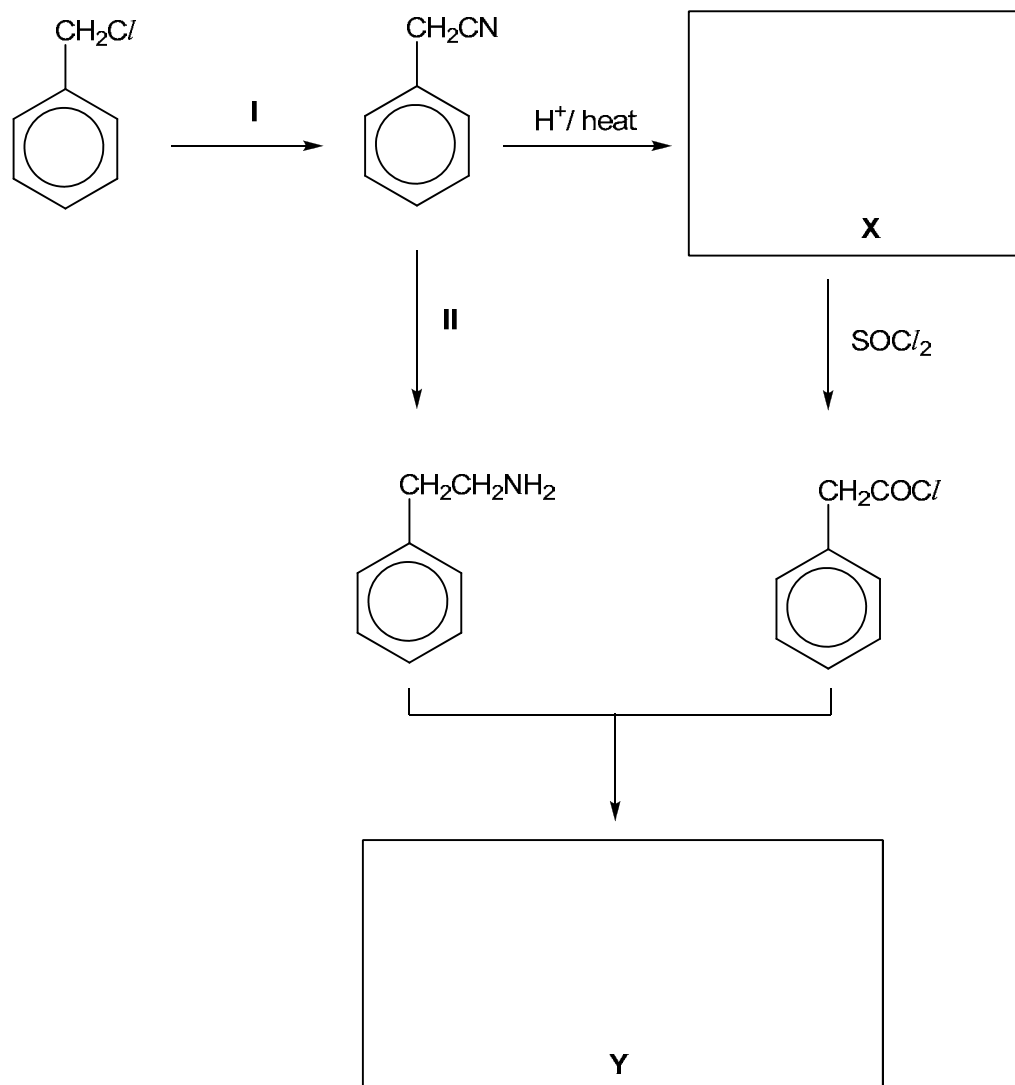
- (e) Give the structural formula of the organic product of reaction 6.



[1]

[Total: 13]

- 6 Compound Y is a useful intermediate in the synthesis of pharmaceuticals. Y can be synthesised from chloromethylbenzene according to the following scheme.



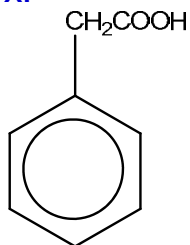
(a) Suggest laboratory reagents and conditions for steps I and II.

		Step I: Reagent: Ethanolic KCN Condition: Heat Step II: Reagent: LiAlH_4 in dry ether Condition: Room Temperature Or NaBH_4 in alkaline methanol
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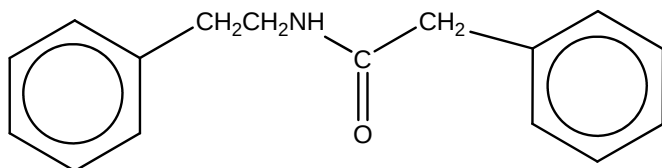
[2]

(b) Draw the structures of **X** and **Y** in the space provided.

X:



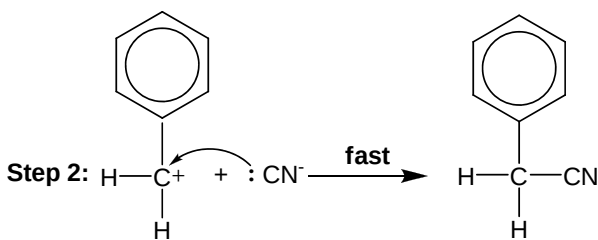
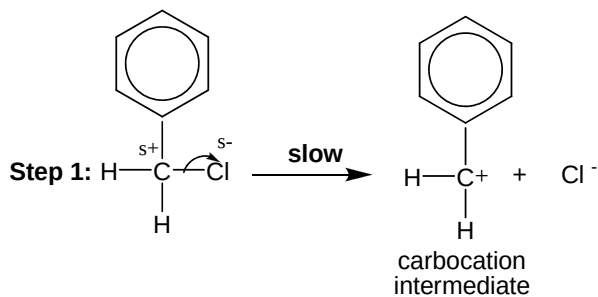
Y:



[2]

(c) Describe the mechanism of step I.

Nucleophilic Substitution ($\text{S}_{\text{N}}1$), also accept $\text{S}_{\text{N}}2$



[3]

(d) Suggest an explanation for the following.

When treated with warm aqueous sodium hydroxide followed by an excess of dilute nitric acid and aqueous silver nitrate, bromoethane gives a cream precipitate but CH_2CHBr gives no precipitate.

$\begin{array}{c} \text{H} \quad \text{H} \\ | \quad | \\ \text{H}-\text{C}=\text{C}-\text{Cl} \end{array}$
 does not undergo nucleophilic substitution with NaOH (aq) as the $\text{C}-\text{Cl}$ bond is strengthened by the overlapping of the p-orbital of Cl with the π orbitals of carbon-carbon double bond / lone pair of electrons on Cl delocalise into benzene ring. Thus the $\text{C}-\text{Cl}$ bond is too strong to be broken. Since there are no free Cl^- ions, no precipitate was formed upon the addition of AgNO_3 (aq).

[2]

(e) Reaction of bromoethane with ethanolic ammonia produces a by-product with the formula $\text{C}_8\text{H}_{20}\text{NBr}$ which gives an immediate precipitate with aqueous silver nitrate. Suggest a structure for this compound and explain why this by-product is formed.

$(\text{CH}_3\text{CH}_2)_4\text{N}^+ \text{Br}^-$ or $(\text{CH}_3\text{CH}_2)_4\text{N Br}$

The amine formed is **more nucleophilic** than ammonia as the **electron-donating alkyl groups** makes the **lone pair of electrons on N atom more readily available for nucleophilic attack** on the electron deficient carbon atom. Therefore, **multiple substitution** occurred.

[2]

[Total: 11]